

PLSC 30600

Week 1 mini deck: Bounded outcomes and missingness

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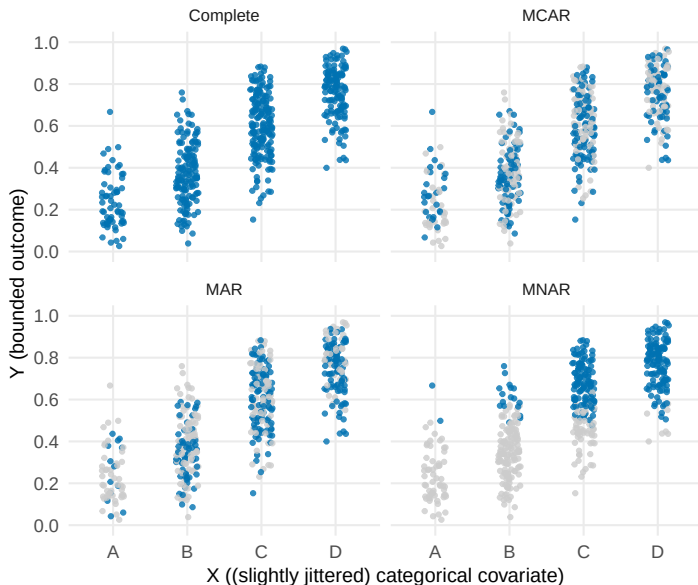
Data generating process (bounded outcomes)

```
set.seed(30600)
n <- 500
X <- factor(sample(c("A","B","C","D"), n, replace = TRUE,
                  prob = c(0.15, 0.25, 0.35, 0.25)))
mu <- c(A = 0.25, B = 0.40, C = 0.60, D = 0.75)[X]
phi <- 10                                # concentration
Y <- rbeta(n, mu * phi, (1 - mu) * phi)

# Missingness mechanisms (50% observed each)
# MCAR:  random half observed
# MAR:   Pr(R=1|X) piecewise by X bins (all X have positive chance)
# MNAR:  selection by Y (higher Y more likely observed)
```

Complete vs. MCAR vs. MAR vs. MNAR

Different missingness mechanisms



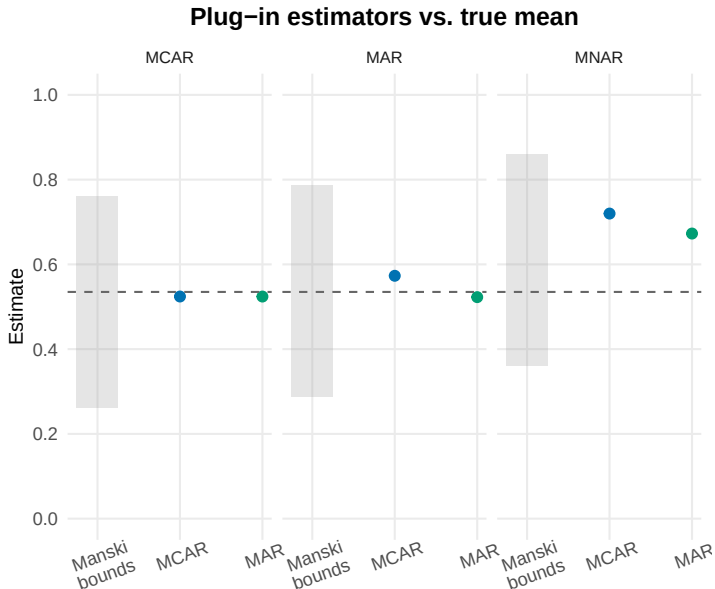
Estimators

```
manski_bounds <- function(y, r) {  
  lower <- mean(ifelse(r == 1, y, 0))  
  upper <- mean(ifelse(r == 1, y, 1))  
  c(lower = lower, upper = upper)  
}
```

```
mu_mcar <- function(y, r) {  
  mean(y[r == 1])  
}
```

```
mu_mar <- function(y, r, x) {  
  x <- factor(x)  
  bin_means <- tapply(y[r == 1], x[r == 1], mean)  
  bin_weights <- prop.table(table(x))  
  sum(bin_means[names(bin_weights)] * bin_weights)  
}
```

Estimators relative to the true mean

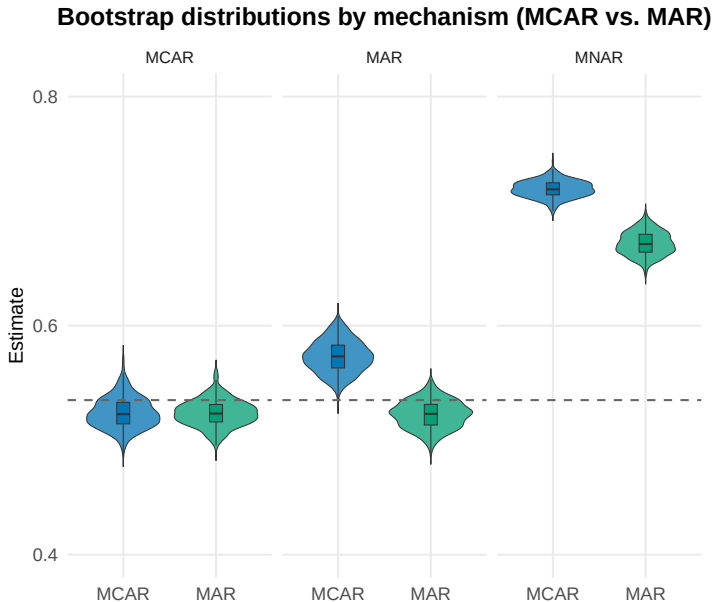


Bootstrap: sampling variability

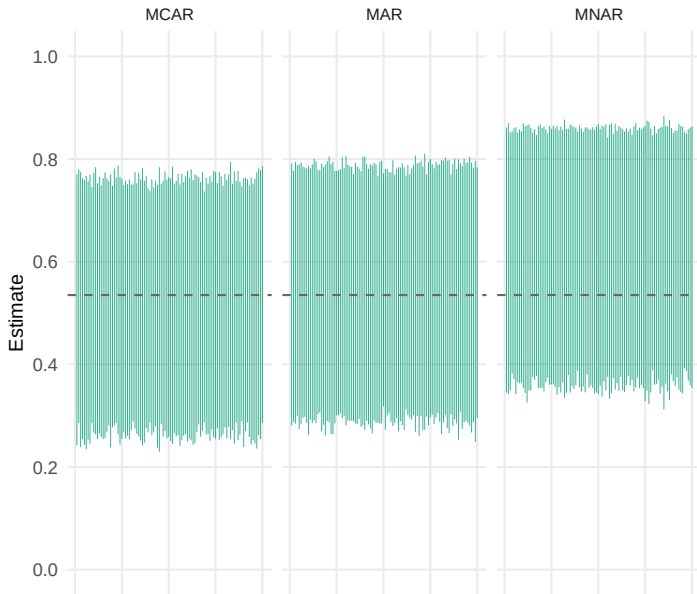
```
B <- 500
boot_df <- lapply(c("MCAR","MAR","MNAR"), function(m) {
  r_all <- list(MCAR = r_mcar, MAR = r_mar, MNAR = r_mnar)[[m]]
  boot_mat <- replicate(B, {
    idx <- sample.int(n, n, replace = TRUE)
    y_b <- Y[idx]
    r_b <- r_all[idx]
    x_b <- X[idx]
    c(mcar = mu_mcar(y_b, r_b),
      mar = mu_mar(y_b, r_b, x_b),
      manski_lo = unname(manski_bounds(y_b, r_b)["lower"]),
      manski_hi = unname(manski_bounds(y_b, r_b)["upper"]))
  })
  as_tibble(t(boot_mat)) |>
  mutate(mechanism = m, iter = row_number())
}) |>
  bind_rows()

boot_long <- boot_df |>
  pivot_longer(cols = c(mcar, mar, manski_lo, manski_hi),
    names_to = "estimator", values_to = "estimate")
```

Bootstrap distributions (MCAR vs. MAR)



Manski bounds



Bootstrap percentile CIs (95%)

```
ci_95 <- boot_df |>
  pivot_longer(cols = c(mcar, mar),
               names_to = "estimator", values_to = "estimate") |>
  group_by(mechanism, estimator) |>
  summarize(
    lo = quantile(estimate, 0.025, na.rm = TRUE),
    hi = quantile(estimate, 0.975, na.rm = TRUE),
    .groups = "drop"
  )
```

Bootstrap summaries + MC CI coverage

Mech.	Est.	CI lo (1 samp)	CI hi (1 samp)	Bias (1 samp)	Cover (MC)
MCAR	MCAR	0.501	0.553	-0.011	0.938
MCAR	MAR	0.502	0.547	-0.011	0.900
MCAR	Manski lo	0.237	0.289	--	--
MCAR	Manski hi	0.737	0.786	--	--
MAR	MCAR	0.548	0.600	0.038	0.062
MAR	MAR	0.499	0.545	-0.013	0.925
MAR	Manski lo	0.257	0.312	--	--
MAR	Manski hi	0.764	0.809	--	--
MNAR	MCAR	0.704	0.734	0.184	0.000
MNAR	MAR	0.654	0.691	0.137	NA
MNAR	Manski lo	0.328	0.389	--	--
MNAR	Manski hi	0.845	0.875	--	--

References I

- Hernán, M. A. and Robins, J. M. (2010). *Causal inference: What if?* CRC Boca Raton, FL.
- Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*, 81(396):945–960.
- Manski, C. F. (2003). Identification problems in the social sciences and everyday life. *Southern Economic Journal*, 70(1):11.
- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3):581–592.